

Dynamic probabilistic risk assessment for electric grid cybersecurity

Xiaoxu Diao^{a,*}, Yunfei Zhao^a, Carol Smidts^a, Pavan Kumar Vaddi^a, Ruixuan Li^b, Hangtian Lei^c, Yacine Chakhchoukh^c, Brian Johnson^c, Katya Le Blanc^b

^a The Ohio State University, OH, United States

^b Idaho National Laboratory, ID, United States

^c University of Idaho, ID, United States

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ABSTRACT

Electric grid cybersecurity risk has become a significant concern of industries and governments. This paper proposes a dynamic probabilistic risk assessment method for electric grid cybersecurity risk analysis. The proposed method helps reduce the reliance on expert judgment, capture a broad range of components and system dynamics, and model the interactions between various contributing entities (e.g., attacker, operator). In addition, the scenarios with multiple events, such as the occurrence of both cyberattacks and failures of physical components, the occurrence of both cyberattacks and operators' (in)correct reactions, are considered and analyzed. For each cyberattack scenario, Monte Carlo simulations are used to obtain possible sequences of the system's evolution under study and then derive risk estimates. As an application of the proposed method, the risk assessment method serves as the basis of risk-informed defense resource allocation to improve electric grid cybersecurity. The proposed method is verified using the IEEE 14-bus system by evaluating different security resource allocations for selected cyberattack scenarios.

1. Introduction

The energy sector, which consists of electricity, oil, nuclear, natural gas, and renewables, is one of the 16 critical infrastructure sectors in the U.S. Electric grids that generate, transmit, and deliver electricity are vital to national interest and security since many other sectors, such as healthcare and financial systems, rely on their reliable operation. The U.S. electric grid is increasing in both size and complexity. The increase in size is due to an increased number of systems and components to meet the increased energy needs. The increase in complexity is due to the introduction of more heterogeneous systems, particularly digital systems used for communication, computation, and control, and the increasing interconnection and dependency between the systems in a grid. As electric grids grow larger and more complex and use more digital systems, unexpected vulnerabilities, for example, an unmonitored remote access point, may be introduced into the grids, and malicious attackers can take advantage of such vulnerabilities to realize physical consequences via cyber means. As noted in the U.S. electric sector cyber threat and vulnerability analysis report [1], the likelihood of cyberattacks against U.S. electric utilities is increasing. Although no cyberattack on U.S. utilities that led to lasting damage has been reported

publicly, the cyberattack on Ukraine's electric power systems in 2015 caused the loss of power to about 225,000 residents for several hours [2]. Such an event or an even more severe one may occur to the U.S. utilities. As a result, the electricity industry is paying increasing attention to the potential issues that cybersecurity events may cause. Risk assessment for cybersecurity events is required for foreseeable consequences caused by cyberattacks and for evaluating the effectiveness of various cyber defense mechanisms.

However, cybersecurity risk assessment for modern grid systems is a challenging task since the grid systems usually include various types of components, such as smart meters, circuit breakers, human-machine interfaces (HMIs), remote terminal units (RTUs), power generation and distribution units, power consumers, regional control centers [3]. The propagation of impacts caused by cyberattacks involves various interacting systems, components, and human operators (e.g., smart meters, protection systems, and on-site operators). The attackers attempt to compromise and control digital systems or networks to increase their negative impacts on the electric grid. On the other hand, defense mechanisms (e.g., protection systems, responsible system operators) respond to intrusions with varying effectiveness to isolate the cyberattacks or mitigate their negative impacts.

This paper proposes a constrained Dynamic Probabilistic Risk

* Corresponding author.

E-mail address: diao.38@osu.edu (X. Diao).

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Nomenclature

$[0, T]$	The time period of risk analysis.	cyberattack	
A_k	A set of cyberattack Actions A that can be applied to the vulnerability V_k .	PW_{branch}	The branch powers
AC_{branch}	Actions to change the state of Branches.	PW_{bus}	The bus powers
AC_{bus}	Actions to change the state of buses;	PW_{gen}	The generator powers
AC_{gen}	Actions to change the power of Generators	PW_{load}	The load powers
AC_{load}	Actions to change the power of Loads	R_{sys}	The system risk.
$B(\cdot)$	A Boolean expression.	S	The set of system states.
BI	The branch currents in the grid system.	$\bar{s}_i$	A vector of states of the components composing the grid system
BN_{open}	The number of open branches.	ST_a	The set of all events caused by cyberattacks.
BV	The bus voltages in the grid system.	ST_{branch}	The branch states
$\mathbb{C} = \{C_1, \dots, C_{N_C}\}$	A set of system components under the configuration CF .	ST_{bus}	The bus states
CF	A system configuration.	ST_e	The set of all events caused by physical faults.
CS_1, CS_2, CS_3	A set of constraints CS_1, CS_2, CS_3 defined for risk analysis.	ST_h	The set of all human actions.
$\mathbb{E} = \{E_1, \dots, E_{N_E}\}$	A set of physical faults in the system under analysis.	$\mathbb{V} = \{V_1, \dots, V_{N_V}\}$	A set of vulnerabilities in the system under analysis under the configuration CF .
$fa_{k,l}(t)$	The distribution over time representing the time when the attack action is conducted.	V_k	A vulnerability in the system under analysis.
$ff_k(t)$	The distribution over time determines the time when the fault will occur.	$\bar{x}$	A vector of physical variables
$fh_k(t)$	The distribution determines the time when the action will be performed.	$X_{k,l}$	A set of the actual actions that can be conducted by attackers.
$f_i(\bar{x}, t)$	The differential equations modeling the evolution of a grid system.	Y_k	A set of behaviors of the target component or link when a physical fault is triggered.
FR_{branch}	The estimated branch frequency	Z_k	The set of human behaviors
$g_i(\bar{y}, t)$	The solution of the differential equations modeling the evolution of a grid system.	$\lambda_i(\bar{x})\Delta t$	The conditional probability of leaving i in the interval Δt .
$g_i^{-1}(\bar{x}, t)$	The inverse function of $g_i(\bar{y}, t)$.	$\delta(\cdot)$	The Dirac delta function
$\mathbb{H} = \{H_1, \dots, H_{N_H}\}$	A set of human reactions.	$\pi(\bar{x}, i, t)$	The probability density of the overall system state
$\mathbb{I}_C = \{IC_1, \dots, IC_{N_C}\}$	A set of initial states of components.	Abbreviation	
$\mathbb{I}_L = \{IL_1, \dots, IL_{N_L}\}$	A set of initial states of links.	AEP	American Electric Power
i, j, q	The index represents a state vector.	API	Application Interface
$\mathbb{L} = \{L_1, \dots, L_{N_L}\}$	A set of links under the configuration CF .	CCE	Cyber informed Engineering
LL	The total loss of loads	CERT	Cyber Emergency Response Team
$\mathcal{M}_A$	The superset of all the possible attack actions.	CVE	Cybersecurity Vulnerabilities Database
N_C	The total number of system components.	CVSS	Common Vulnerability Scoring System
N_E	The total number of physical faults.	DOE	Department of Energy
N_{fail}	The number of simulations in which at least one of system failures occurred.	DPRA	Dynamic Probabilistic Risk Assessment
N_H	The total number of considered human actions.	FDI	False Data Injection
N_L	The total number of links.	FMECA	Failure Mode Effects and Criticality Analysis
N_{sim}	The total runs of simulations.	FPBN	Fuzzy Probability Bayesian Network
N_V	The total number of vulnerabilities.	HMI	Human-Machine Interface
$p(i \rightarrow j \bar{u})\Delta t$	The conditional probability that the state i has a transition to j in the interval Δt when the state vector is $\bar{u}$.	ICS	Industrial Control System
$pa_{k,l}$	The probability that the attacker will choose the action $X_{k,l}$ given that the vulnerability V_k is utilized	IEEE	Institute of Electrical and Electronics Engineers
pf_k	The probability of the physical fault occurrence.	INL	Idaho National Laboratory
ph_k	The success probability of human actions	IT	Information Technology
$ps_{k,l}$	The probability that the action set $X_{k,l}$ can be successfully conducted by utilizing the vulnerability V_k .	LDRD	Laboratory Directed Research & Development
pv_k	The probability that the vulnerability V_k will be chosen for	MDP	Markov Decision Process
		MITM	Man-in-the middle
		MW	Mega Watt
		NIST	National Institute of Standards and Technology
		PMU	Phasor Measurement Unit
		RTU	Remote Terminal Unit
		SCADA	Supervisory Control and Data Acquisition
		SGAM	Smart Grid Architecture Model
		SQL	Structured Query Language
		SUA	System Under Analysis

Assessment (DPRA) method to overcome the challenges in analyzing cybersecurity (e.g., the variety of related components, the attack actions conducted by the attackers, the interactions between the attackers and defense mechanisms, etc.). Specifically, this paper develops a DPRA method to analyze system risk as it relates to cybersecurity issues that can affect electric grid systems. This method considers uncertainties in

the behaviors of attackers, such as the uncertainties in detecting and utilizing a vulnerability in the target system and the uncertainty in successfully compromising a component, and the uncertainties in the responses of cyberattack countermeasures deployed within the grid, such as the uncertainty in the response of the protection systems to mitigate the impact of a cyberattack, the uncertainty in the behaviors of

the system operators under various conditions, and the uncertainty in the occurrence of component failures due to natural causes (i.e., not caused by cyberattacks). Some of these uncertainties are affected by the runtime states of the grid and will significantly influence the consequences of cyber-attacks. The method provides an elegant way of modeling these uncertainties and the complex interactions and dependencies between various pertinent entities, including physical systems, digital systems, and human operators. Also, conditions related to those uncertainties can be defined as constraints for cybersecurity risk analysis to accelerate the analysis process. For example, to analyze the system risk under the condition that the attacker has compromised one of the system components, we can define the probability that the attacker can successfully exploit a vulnerability and compromise the related component as 100%. During the analysis, the simulation process will use those probabilities, and the analysis results related to those predefined conditions (i.e., constraints) can be obtained. As a case study system, an IEEE 14-bus system is used to verify the proposed method by evaluating several defense strategies under various cyberattack scenarios. The modules and scripts developed in the case study analysis can be modified with little effort and reused in the analysis for other electric grids.

In the rest of the paper, Section 2 reviews existing methods for cybersecurity risk assessment. Section 3 details the proposed DPRA method. Section 4 applies the proposed method to the IEEE 14-bus system as a case study. Section 5 discusses the limitation and generalizations of the proposed method. Finally, Section 6 concludes this paper and discusses future research.

2. Related work

In recent years, a significant amount of attention has been paid to cybersecurity risk assessment. For example, the Common Vulnerability Scoring System (CVSS), which defines qualitative metrics for risk assessment specific to Information Technology (I.T.) systems, was utilized to measure the security level of computer systems [4]. The Consequence-driven Cyber-informed Engineering (CCE) method [5,6] examines the industrial control systems (ICS) environment for high-impact events/risks. In this method, the analyst needs to assign values to the area impacted, attack duration, attack breadth, etc., in determining the consequence of a cyber event based on his/her subjective judgment. A cybersecurity risk assessment method based on the Smart Grid Architecture Model (SGAM) was developed in [7] to obtain the potential risk for each element in the model by rating the probability of occurrence and the impact of a successful attack. Another risk assessment framework was developed in [8] which connects the hierarchical holographic model with the risk filtering and ranking model and the failure mode, effects and criticality analysis (FMECA).

However, those methods heavily rely on expert judgment and experience. Overreliance on expert judgment requires analysts to possess a high level of expertise and may also lead to reproducibility issues. Other studies have used fuzzy logic or Bayesian network methods to assist experts in modeling system dependencies and improving the assessment. For example, Peng et al. [9] applied a Bayesian network model to optimize an online data-driven parameter learning strategy which can improve the accuracy of dynamic assessment results. Demertzis et al. [10] employed soft computing approaches, fuzzy cognitive maps, and a Mamdani fuzzy inference system to model the overall security level. Zhang et al. [11] proposed a fuzzified probability Bayesian network (FPBN) approach to analyze and predict the propagation of impacts caused by cyberattacks. They also developed an approximate dynamic inference algorithm for the dynamic assessment of ICS cybersecurity risk. Nevertheless, system dynamics and attackers' behaviors are not considered explicitly in these studies. Woo et al. [12] developed a model-based assessment framework to perform qualitative evaluations for cyber treats and vulnerabilities for the Supervisory Control and Data Acquisition (SCADA) systems.

As noted by Song et al. [13], system dynamics are important in determining the consequences of an abnormal event and should be considered in risk assessment for electric power systems. In order to overcome these shortcomings, Chen et al. [14] proposed a behavior model to investigate the risk of cyberattacks on power grids. The model considers the attacker's behavior, including attack resource allocation and attack target selection, by using the probability of a successful response and utility attenuation model based on analysis of historical events. Wang et al. [15] analyzed the impacts of cyberattacks targeting the communication channels between sensors and the control center. Liu et al. [16] proposed a method to assess risks caused by malicious attacks on protection system settings and parameters. Zhang et al. [17] proposed a framework for assessing the robustness of cyber-coupled power grids during cascading failures. The framework considered the physical layer, the control systems, and the protection systems. Ding et al. [18, 19] developed a probabilistic model to assess the unobservable risk of the power grid when the Phasor measurement units (PMUs), the device integrated into the smart grid for monitoring the operational state of a system and improving the reliability, are compromised by cyberattacks. However, these studies focus on limited scopes of cybersecurity risk analysis, for example, specific components, and do not provide a comprehensive approach to cybersecurity risk analysis.

Besides these approaches, methods for analyzing more complex scenarios, which include the competition between cyber attackers and system defenders, have been proposed recently, such as the ones using game theory [20–23], the Markov Decision Process (MDP) [24,25], Attack Graphs [26], and Petri-nets [27]. However, these methods do not explicitly model the behaviors of the intelligent sensors, protection systems, and the state estimator, all of which play essential roles in the evolution of a cyberattack scenario.

Probabilistic risk assessment provides a systematic approach for quantifying the risk to a system by analyzing all credible scenarios in which a system can be disrupted, the likelihood of each scenario, and the consequence of each scenario. Probabilistic risk assessment has been widely used in various fields, including nuclear [24], aerospace [28], oil and gas [29], etc. Existing probabilistic risk assessment efforts have focused on disturbances caused by natural events, for example, pump failures, floods, etc. This research proposes a Dynamic Probabilistic Risk Assessment (DPRA) simulation environment for electric grid cybersecurity risk assessment. DPRA has been applied to probabilistic risk assessment for cascading events in electric grids, for example, for the slow cascade phase and the fast cascade phase in [30–32], but these studies have focused on risk related to system failures due to thermal transients instead of system compromises caused by cyberattacks. DPRA has seen applications in various areas, in particular the nuclear industry [33–36].

This paper introduces a DPRA method that considers various actors and system components that may contribute to the consequences of a cyberattack, such as the attackers who intend to compromise the electric grid, the system operators who are responsible for mitigating the impacts of cyberattacks, the protection system which includes many circuit breakers to prevent damages, and the state estimator which support system state diagnosis. Also, before conducting the risk assessment, knowledge related to the attackers, the defense mechanism, or the electric grid can be defined as constraints to accelerate the DPRA process and improve the analysis results. None of the existing methods can perform such analyses.

3. DPRA methodology

3.1. Overview

An electric grid is a complex cyber-physical system. An incident in an electric grid caused by a cyberattack may evolve rapidly and the compromises of systems in the grid and cascading failures may lead to catastrophic consequences. Compared to other cyber-physical systems,

an electric grid has the following characteristics:

- High complexity resulting from complex communications and interactions. The complexity includes not only the topology of the communications between various components of the grid but also the complexity due to the interactions between the components and actors. Faithfully representing the architecture of the system and the interactions between various components and actors is critical for accurately assessing the risk.
- Wide range of uncertainty. Since the responses of various entities (the attacker, the system operator, the protection systems, the state estimator, a physical component) have a direct impact on the consequences of cyberattacks, a wide range of uncertainties need to be considered in the risk assessment methodology.

This paper proposes a dynamic PRA method for electric grid cybersecurity risk analysis. The proposed method has the following capabilities. First, a grid system simulator with physics-based models is used to simulate the trajectories of system state evolutions. The simulator is used to calculate system dynamics under various cyberattacks and component failures following the cyberattacks. Second, dynamic interactions between system components (digital and physical) are modeled and simulated. The behaviors of these components (e.g., the

sensors, the breakers, and the protection system) are both time-dependent and physics dependent, and they have a significant impact on the consequences of cyberattacks against the grid. Third, human interactions with the grid system are considered. Classical reactions of system operators under various conditions are modeled and simulated depending on physics, time, and states of other components. These reactions will also dynamically change the trajectories of system state evolutions. For example, manually opening the circuit breaker of a faulty branch will isolate the grid from a short-to-ground fault and therefore change the evolution of the incident. Fourth, stochastic component failures caused by natural degradations are also considered in the risk assessment. Such failures may amplify the impact of cyberattacks and may neutralize the mitigation actions carried out by the protection system or the system operators. Table 1 shows the uncertainties considered in the proposed methodology.

Fig. 1 provides further details of the proposed DPRA methodology for assessing the impact of cyberattacks on electric grids. In the simulation, multiple functional modules are used to simulate the behavior of their corresponding components in the grid system. This includes the responses of physical components (e.g., branches, buses modeled by the Physics Module), protection systems, the state estimator, the HMI, and the operators. The responses of physical components are modeled by a simulator of the electric grid in which the physical components'

Table 1
Uncertainties considered in the proposed methodology.

Module Types & Links	Uncertainties
Cyber Module & Link	• Types of vulnerabilities; • Attack actions that can be conducted; • Whether or not a vulnerability will be exploited by the attacker; • Whether or not an attack action will be used by the attacker; • Whether or not an attack action will be effective in compromising a component; • When an attack will be initiated in a scenario;
Physical Module	• The occurrence of a physical failure in a scenario; • When the failure will occur.
Human Module	• The system operator's response to an incident/cyberattack, i.e., success or failure. • When the operator will respond.

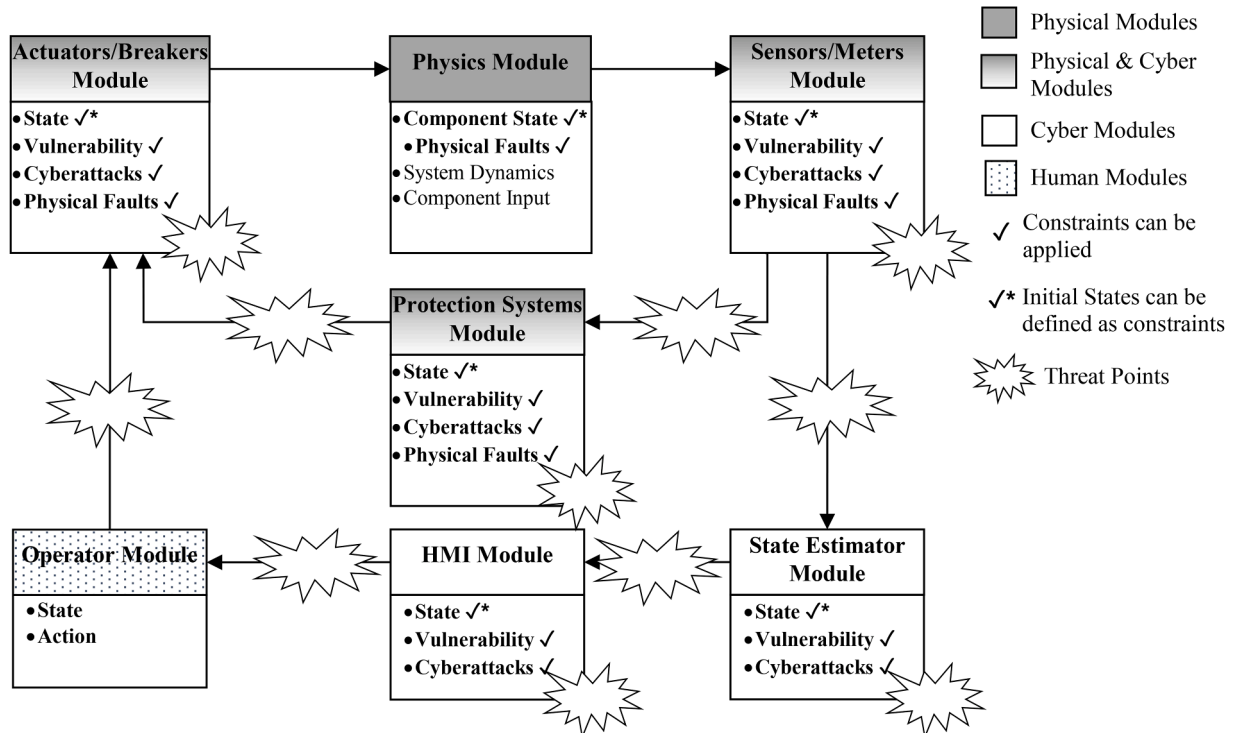


Fig. 1. Diagram of the constrained DPRA methodology.

behaviors (buses, branches, generating stations etc.) depend not only on their states but also on their inputs which are derived from the physical components that are linked to them (e.g., a bus may be linked to one or more branches). The responses of protection systems, the state estimator, the HMI, and the operators are modeled by the functional modules developed in the DPRA method.

In the figure, the background color of each block, representing a module, denotes the types (e.g., cyber or physical) of the systems in the module. For example, the white color denotes modules for cyber systems and the gray color denotes modules for physical systems. The gradient background from gray to white means that the module consists of both cyber and physical systems. Each box in the figure also lists the characteristics that can be modeled by the proposed method for the represented module. The characteristics with a correct mark accept predefined conditions in the cybersecurity risk analysis, i.e., those characteristics can serve as the constraints for cybersecurity risk analysis. As an example, digital technology has been used in protection systems. Assume that a known vulnerability has been verified and is existing in the current version of the protection system. Then, the probability of such a vulnerability for the protection system can be predefined as 100%, and the systems are susceptible to cyberattacks (shown in the middle of the figure). During the analysis, those probabilities will be used by the simulation process and improve the analysis results. Therefore, the proposed method is defined as a constrained DPRA method.

In the figure, the links and arrows denote the data flow in the cybersecurity risk analysis. For example, operators obtain information

on the system state from the HMI, the information displayed on which in turn is from the state estimator. Operators then take actions that influence system dynamics, modeled in the physics module.

Table 2 shows the input and output variables of each module in the proposed method. Due to the connection relations between the modules, the outputs of some modules will be mapped to the inputs of some other modules. Therefore, the interactions between these modules are defined by the mapping relations.

3.2. Formalization

In this section, we formalize the proposed DPRA method for cybersecurity risk analysis using the theory of continuous-time event trees first introduced in [34].

In a grid system, the physical state can be defined as a vector of physical variables $\bar{x}$, which is a vector of physical variables, such as the voltages of buses, the currents on branches, etc., where $\bar{x} \in \mathbb{R}^n$. Also, the system states $S = \{\bar{s}_1, \dots, \bar{s}_n\}$ can be defined as a set of vectors in which $\bar{s}_i$ is a vector of states of the components composing the grid system. In this paper, these components can be the physical components (e.g., buses or generators), the cyber components (e.g., the smart sensors and the HMI), or the operators (since we model the operator as a part of the grid system). The possible states of a physical component can be nominal or faulty. The possible states of a cyber component can be nominal or compromised. And the operator states can be successful (in which the operator correctly responds to an incident) or failed (in which the operator does not respond or incorrectly responds to an incident). In this

Table 2
Interaction and interdependency variables in the proposed method.

Variables	Outputs from the following Modules	Inputs to the following Modules
• ST_{branch}, Branch States** • ST'_{branch}, Measured Branch States** • ST'^i_{branch}, Estimated Branches States**	System Physics Sensors/Meters* State Estimator* HMI*	Sensors/Meters Protection Systems HMI Operator
• ST_{bus}, Bus States** • ST'_{bus}, Measured Bus States** • ST'^i_{bus}, Estimated Bus States**	System Physics Sensors/Meters* State Estimator* HMI*	Sensors/Meters Protection Systems HMI Operator
• BV, Bus Voltages • BV', Measured Bus Voltages**	System Physics Sensors/Meters*	Sensors/Meters State Estimator Protection Systems
• BV', Estimated Bus Voltages**	State Estimator* HMI*	HMI Operator
• BI, Branch Currents • BI', Measured Branch Currents**	System Physics Sensors/Meters*	Sensors/Meters State Estimator Protection Systems
• BI', Estimated Branch Currents*	State Estimator* HMI*	HMI Operator
• PW_{bus}, Bus Powers • PW'_{bus}, Measured Bus Powers** • PW'^i_{bus}, Estimated Bus Powers**	System Physics Sensors/Meters* State Estimator* HMI*	Sensors/Meters State Estimator HMI Operator
• PW_{branch}, Branch Powers • PW'_{branch}, Measured Branch Powers** • PW'^i_{branch}, Estimated Branch Powers**	System Physics Sensors/Meters* State Estimator* HMI*	Sensors/Meters State Estimator HMI Operator
• FR'_{branch}, Estimated Branch Frequency**	State Estimator* HMI*	HMI Operator
• AC_{branch}, Actions to change the states of Branches**	Protection Systems* Operator	Actuator/Breaker
• AC_{bus}, Actions to change the states of buses**	Protection Systems* Operator	Actuator/Breaker
• AC_{gen}, Actions to change the powers of Generators • AC_{load}, Actions to change the powers of Loads	Operator Operator	Actuator/Breaker Actuator/Breaker
• ST^+_{branch}, Branch States for the next time step • ST^+_{bus}, Bus States for the next time step • PW^+_{gen}, Generator Powers for the next time step • PW^+_{load}, Load Powers for the next time step	Actuator/Breaker* Actuator/Breaker* Actuator/Breaker* Actuator/Breaker*	System Physics System Physics System Physics System Physics

* Variables that can be compromised (modified) by the attacker when the corresponding components are compromised.

** Variables that can be compromised (modified) by the attacker when the networks for transferring the value of such variables are compromised.

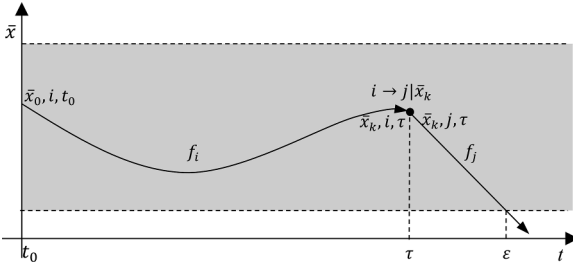


Fig. 2. A possible trajectory of physical variables.

paper, we define ST_e as the set of all events caused by physical faults in a grid system, define ST_a as the set of all events caused by cyberattacks and define ST_h as the set of all events caused by human actions. Also, we use the index i to represent the state vector $\bar{s}_i$. Then, the overall state of the grid system at a specific time can be denoted by a tuple $(\bar{x}, i, t)$. The trajectory of the evolution of a grid system can be modeled by differential equations, as presented by Eq. (1).

$$\frac{d\bar{x}}{dt} = f_i(\bar{x}, t) \quad (1)$$

Fig. 2 shows a possible trajectory of $\bar{x}$. In the figure, we define that the initial condition of the trajectory is $(\bar{x}_0, i, t_0)$. From that condition, the system evolves along the curve of the function f_i . Then, a change of the system state occurred $(i \rightarrow j | \bar{x}_k)$ at time τ . We can see that the differential function of system evolution changes from f_i to f_j correspondingly. Assume that the gray box defines the safe zone of the trajectory, i.e., the state of the grid system will be unsafe (or failed) when the trajectory of $\bar{x}$ exits the gray box. From the figure, we can observe that the trajectory exceeds the boundary of the safe zone at time ϵ . In this paper, one of the objectives of the proposed method is to obtain the probability that the trajectory of $\bar{x}$ exits the safety boundary.

Assume that we can obtain the solution of Eq. (1) as g_i . Therefore,

$$\bar{x}(t) = g_i(\bar{y}, t) \quad (2)$$

Using $\pi(\bar{x}, i, t)$ as the probability density of the overall system state, if we assume that the grid system we are investigating is Markovian, i.e., the state of the grid system at time $t + dt$ is only conditioned by its state at time t , based on the equations proved by [34], we can learn that

$$\begin{aligned} \pi(\bar{x}, i, t) = & \int \pi(\bar{u}, i, 0) \delta[\bar{u} - g_i^{-1}(\bar{x}, t)] d\bar{u} \\ & + \sum_{j \neq i} \int p(j \rightarrow i | \bar{u}) \int \pi(\bar{u}, j, \tau) \delta[\bar{u} - g_i^{-1}(\bar{x}, t - \tau)] d\tau d\bar{u} \end{aligned} \quad (3)$$

Where,

- $g_i^{-1}(\bar{x}, t)$ is the inverse function of $g_i(\bar{y}, t)$, hence, if $g_i^{-1}(\bar{x}, t) = \bar{y}$ then $g_i(\bar{y}, t) = \bar{x}$;
- $\delta[\bar{u} - g_i^{-1}(\bar{x}, t)]$ is the Dirac delta function, in which

$$\begin{aligned} \delta[\bar{u} - g_i^{-1}(\bar{x}, t)] & \cong \begin{cases} +\infty, & \bar{u} = g_i^{-1}(\bar{x}, t) \\ 0, & \bar{u} \neq g_i^{-1}(\bar{x}, t) \end{cases} \\ & \text{and } \int_{-\infty}^{+\infty} \delta[\bar{u} - g_i^{-1}(\bar{x}, t)] d\bar{u} = 1 \end{aligned}$$

- $p(i \rightarrow j | \bar{u}) \Delta t$ is the conditional probability that the state i has a transition to j in the interval Δt when the state vector is $\bar{u}$.

In Eq. (3), the first term $\int \pi(\bar{u}, i, 0) \delta[\bar{u} - g_i^{-1}(\bar{x}, t)] d\bar{u}$ denotes the probabilities that the trajectory evolves from all the possible initial states $\pi(\bar{x}_0, i, 0)$ to the current state $\pi(\bar{x}, i, t)$ without a change of states

$(i \rightarrow j)$. The second term $\sum_{j \neq i} \int p(j \rightarrow i | \bar{u}) \int \pi(\bar{u}, j, \tau) \delta[\bar{u} - g_i^{-1}(\bar{x}, t - \tau)] d\tau d\bar{u}$

denotes all the trajectories in which the system state changed from other states j to the current state i .

Generally, in the grid system, a state change $(i \rightarrow j)$ can be caused by an event e of physical faults $i \xrightarrow{e} j$, an event a of cyberattacks $i \xrightarrow{a} j$, or an event h caused by the system operator $i \xrightarrow{h} j$. If we assume that the transitions from nominal states to those abnormal states are independent. Then $p(i \rightarrow j | \bar{x})$ can be extended as:

$$\begin{aligned} p(i \rightarrow j | \bar{x}) = & \sum_{e \in ST_e} p(i \rightarrow j | \bar{x}, e) p(e | \bar{x}) + \sum_{a \in ST_a} p(i \rightarrow j | \bar{x}, a) p(a | \bar{x}) \\ & + \sum_{h \in ST_h} p(i \rightarrow j | \bar{x}, h) p(h | \bar{x}) \end{aligned} \quad (4)$$

To acquire the probability that system trajectories deviate from the safety zone (as the example shown in Fig. 2), we need to calculate the elements in the conditional probabilities $p(i \rightarrow j | \bar{x})$.

Assume that a system configuration CF is a superset of system components and links with their initial conditions, vulnerabilities, and failure modes $CF = \{\mathbb{C}, \mathbb{I}_C, \mathbb{L}, \mathbb{I}_L, \mathbb{V}, \mathbb{E}\}$, where $\mathbb{C}, \mathbb{I}_C, \mathbb{L}, \mathbb{I}_L, \mathbb{V}, \mathbb{E}$ are defined in the remainder of the paragraph. The component set $\mathbb{C} = \{C_1, \dots, C_{N_C}\}$ is defined where C_k denotes the k th component and N_C is the total number of components. $\mathbb{C}$ may include smart sensors, state estimators, or HMIs. The initial state of these component $\mathbb{I}_C = \{IC_1, \dots, IC_{N_C}\}$ can be normal or faulty. The network connections set $\mathbb{L} = \{L_1, \dots, L_{N_L}\}$ is defined where L_k denotes the k th link between the cyber components in the grid system, such as industrial ethernet, and N_L is the total number of links. The initial state of the links $\mathbb{I}_L = \{IL_1, \dots, IL_{N_L}\}$ can be connected or open. Each component or link may possess zero, one, or more vulnerabilities. For convenience, we define $\mathbb{V} = \{V_1, \dots, V_{N_V}\}$ as the set of all the vulnerabilities in the system under analysis, where N_V is the total number of vulnerabilities. Each vulnerability can be mapped to a component or a link. The set of possible physical failure can be defined as $\mathbb{E} = \{E_1, \dots, E_{N_E}\}$, where N_E is the total number of considered failures.

Since every vulnerability in the system under analysis may be exploited by an attacker and used for compromising a component or a link, we can define the vulnerability selection vector $\mathcal{S}_V$ and the super set of all the possible attack actions $\mathcal{A}$ as below.

$$\mathcal{S}_V = \{pv_1(\bar{x}, i), \dots, pv_{N_V}(\bar{x}, i)\} \quad (5)$$

$$\mathcal{A} = \{A_1, \dots, A_{N_V}\} \quad (6)$$

where,

- The probability $pv_k(\bar{x}, i)$ denotes the probability that the vulnerability V_k will be selected for a cyberattack while being in state $(\bar{x}, i)$.
- A_k denotes the set of attack actions that can be conducted by an attacker when the vulnerability V_k is compromised,

Furthermore, the set A_k can be extended as a set of various attack actions with success probability and time distributions.

$$A_k \equiv (X_{k,1}, pa_{k,1}(\bar{x}, i), ps_{k,1}(\bar{x}, i, q), fa_{k,1}(\bar{x}, i, t)), \dots,$$

$$(X_{k,m_k}, pa_{k,m_k}(\bar{x}, i), ps_{k,m_k}(\bar{x}, i, q), fa_{k,m_k}(\bar{x}, i, t)) \quad (7)$$

where,

- The set $X_{k,l}$ represents one or multiple actual actions that can be conducted by the attackers targeting the vulnerability V_k . For example, the set $X_{k,l}$ may include the cyberattack action which changes the sampled voltage of a grid bus from its original value to zero.
- The probability $pa_{k,l}(\bar{x}, i)$ defines the probability that the attacker will choose the action $X_{k,l}$ given that the vulnerability V_k is being selected as a target of attack which the system is in state $(\bar{x}, i)$.
- The probability $ps_{k,l}(\bar{x}, i, q)$ denotes the probability that the action set $X_{k,l}$ can be successfully conducted and change the system state from i to q . The probability may not be 100% since the attacker may not correctly use the attack tool, or the defense mechanism may successfully block the attack.
- The distribution over time $fa_{k,l}(\bar{x}, i, t)$ represents the distribution of the time at which the attack is conducted.

Therefore, the probabilities of state changes caused by cyberattacks can be calculated as below.

$$\sum_{a \in ST_a} p(i \rightarrow j | \bar{x}, a) p(a | \bar{x}) = \sum_{v \in DV} \sum_{q \in S} \sum_{a \in M_A} \sum_{m \in a} \int_0^t p v_v(\bar{x}, i) p a_{a,m}(\bar{x}, i) p s_{a,m}(\bar{x}, i, q) f a_{a,m}(\bar{x}, i, s) \delta_{q,j} ds \quad (8)$$

Where $\delta_{i,j}$ is the Kronecker Delta function.

$$\delta_{q,j} = \begin{cases} 1; & \text{if } q = j \\ 0; & \text{if } q \neq j \end{cases}$$

By defining the parameters in $\mathcal{V}$ and $\mathcal{M}_A$, the proposed method will generate the outputs of cyber components (e.g., smart sensors) and the data transferred through links.

In the proposed method, we introduce components' failure modes caused by physical faults $\mathbb{E} = \{E_1, \dots, E_{N_E}\}$. Each failure mode can be defined as below.

$$E_k = \{Y_k, pf_k(\bar{x}, i, q), ff_k(\bar{x}, i, t)\} \quad (9)$$

where,

- Y_k denotes the physical failure of the target component. For example, when a branch is shorted to ground, the voltage of that branch will become zero.
- The notation $pf_k(\bar{x}, i, q)$ denotes the probability of the failure event which changes the system state from i to q .
- The distribution $ff_k(\bar{x}, i, t)$ determines the time at which the failure will occur.

The probabilities of state changes caused by physical faults can be calculated as below.

$$\sum_{e \in ST_e} p(i \rightarrow j | \bar{x}, e) p(e | \bar{x}) = \sum_{q \in S} \sum_{k \in \mathbb{H}} \int_0^t p f_k(\bar{x}, i, q) f f_k(\bar{x}, i, t) \delta_{q,j} ds \quad (10)$$

In addition, we define human reactions as $\mathbb{H} = \{H_1, \dots, H_{N_H}\}$, where N_H is the total number of human actions considered. Further, we define a human action as

$$H_k = \{Z_k, ph_k(\bar{x}, i, q), fh_k(\bar{x}, i, t)\} \quad (11)$$

where,

- Z_k denotes the set of human behaviors, such as sending a command to open a branch in the grid.
- The notation $ph_k(\bar{x}, i, q)$ denotes the success probability of the human action which changes the system state from i to q .
- The distribution $fh_k(\bar{x}, i, t)$ determines the time at which the action will be completed.

Then, the probabilities of state changes caused by human actions can be calculated.

$$\sum_{h \in ST_h} p(i \rightarrow j | \bar{x}, h) p(h | \bar{x}) = \sum_{q \in S} \sum_{k \in \mathbb{H}} \int_0^t p h_k(\bar{x}, i, q) f h_k(\bar{x}, i, s) \delta_{q,j} ds \quad (12)$$

To calculate the probability of system failure, let $\lambda_i(\bar{x})\Delta t$ be the conditional probability to leave i in the interval Δt and let j be a failure state of the system.

$$\lambda_i(\bar{x}) = \sum_{j \neq i} p(i \rightarrow j | \bar{x}) \quad (13)$$

Then, we can infer that:

$$\lambda_i(\bar{x}) = \sum_{j \neq i} \left[\sum_{e \in ST_e} p(i \rightarrow j | \bar{x}, e) p(e | \bar{x}) \right] + \sum_{j \neq i} \left[\sum_{a \in ST_a} p(i \rightarrow j | \bar{x}, a) p(a | \bar{x}) \right] + \sum_{j \neq i} \left[\sum_{h \in ST_h} p(i \rightarrow j | \bar{x}, h) p(h | \bar{x}) \right] \quad (14)$$

Generally, it is difficult to obtain the inverse function $g_i^{-1}(\bar{x}, t)$. Hence, we use the lemmas below proved by [34] to calculate $\pi(\bar{x}, i, t)$.

$$\delta[\bar{y} - g_i^{-1}(\bar{x}, t)] = \delta[\bar{x} - g_i(\bar{y}, t)] \times \exp \left\{ - \int_0^t \lambda_i[g_i(\bar{y}, s)] ds \right\} \quad (15)$$

Combining the Eqs. (3), (14), and (15), we can obtain that:

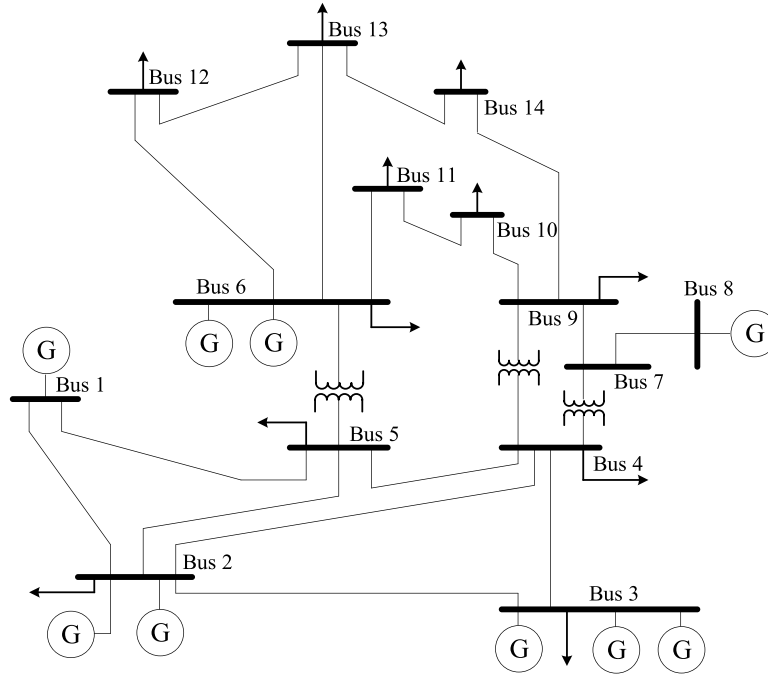


Fig. 3. The diagram of the IEEE 14-bus system.

Table 3

Number of elements analyzed in the IEEE 14-bus system.

Elements	Number	Elements	Number	Elements	Number
Branches	20	Generators	9	Sensors/Meters	270
Bus	14	Loads	11	Breakers	40
State Estimator	1	HMI	1	Operator	1

$$\begin{aligned}
\pi(\bar{x}, i, t) = & \int \pi(\bar{u}, i, 0) \delta[\bar{x} - g_i(\bar{u}, t)] \exp \left\{ - \int_0^t \lambda_i[g_i(\bar{u}, s)] ds \right\} d\bar{u} + \sum_{j \neq i} \sum_{e \in ST_e} \int p(i \rightarrow j | \bar{u}, e) p(e | \bar{x}) \int_0^t \pi(\bar{u}, j, \tau) \delta[\bar{x} - g_i(\bar{u}, t - \tau)] \times \exp \left\{ - \int_0^{t-\tau} \lambda_i[g_i(\bar{u}, s)] ds \right\} d\tau d\bar{u} \\
& + \sum_{j \neq i} \sum_{a \in ST_a} \int p(i \rightarrow j | \bar{u}, a) p(a | \bar{x}) \int_0^t \pi(\bar{u}, j, \tau) \delta[\bar{x} - g_i(\bar{u}, t - \tau)] \times \exp \left\{ - \int_0^{t-\tau} \lambda_i[g_i(\bar{u}, s)] ds \right\} d\tau d\bar{u} + \sum_{j \neq i} \sum_{h \in ST_h} \int p(i \rightarrow j | \bar{u}, h) p(h | \bar{x}) \int_0^t \pi(\bar{u}, j, \tau) \delta[\bar{x} - g_i(\bar{u}, t - \tau)] \times \exp \left\{ - \int_0^{t-\tau} \lambda_i[g_i(\bar{u}, s)] ds \right\} d\tau d\bar{u} \\
& - \int_0^{t-\tau} \lambda_i[g_i(\bar{u}, s)] ds \} d\tau d\bar{u}
\end{aligned} \tag{16}$$

In practice, when the dimension of $\bar{x}$ is larger than 3, due to the overwhelming complexity in solving the Eq. (16), Monte Carlo is adequate for solving this problem [34]. Therefore, in this paper, the system risk, i.e., the estimated probability of system failures, can be calculated by using the ratio between the number of simulations N_{fail} in which at least one of system failures occurred and the total runs of simulations N_{sim} , as shown in Eq. (17).

$$R_{sys} \equiv \frac{N_{fail}}{N_{sim}} \tag{17}$$

In practice, the conditions for identifying system failures can be defined based on the values or functions of physical variables $\bar{x}$. As shown in Fig. 2, these conditions define the safety zone of system trajectories for $\bar{x}$, i.e., the boundary of the gray box. In the proposed method, the physical variables that can be used to define such boundaries include branch states ST_{branch} , bus states ST_{bus} , bus voltages BV ,

branch currents BI , and the powers of buses PW_{bus} , of branches PW_{branch} , of generators PW_{gen} , and of loads PW_{load} . In addition, we can introduce constraints into the simulation to accelerate the evaluation process. In practice, we may ensure that at least one cyberattack will occur in each scenario by adjusting the vulnerability selection vector $\mathcal{D}_V$ as

$$CS_1 \equiv pv_1 + \dots + pv_{N_V} = 1, \quad pv_k \in \mathcal{D}_V \tag{18}$$

Also, for each A_k in $\mathcal{A}_A$, we can ensure that

$$CS_2 \equiv pa_{k,1} + \dots + pa_{k,m_k} = 1 \tag{19}$$

The physical failure probabilities can also be configured as below to evaluate the defense strategies when physical faults occur during cyberattacks,

$$CS_3 \equiv pf_1 + \dots + pf_{N_E} = 1, \quad pf_k \in E_k \tag{20}$$

It is worth noting that by introducing the constraints above to the Monte Carlo simulations, the system risk R_{sys} in Eq. (17) becomes a

Table 4

Module Configuration of the functional modules.

Modules	Configurations
System Physics Module Actuator/Breakers Module	- Since the Power World simulator is used, the simulator's project file implements the configurations for the physics model. - The time delay for a response is zero.
Fault Injection Module	The fault generation module, which deliberately generates branch or bus faults, is designed to analyze cybersecurity risk when physical faults as also considered in the analysis. No fault will be generated for the nominal and cyber-only scenarios. The "three-phase short to ground" fault will be injected into the system with 100% probability at time 1.0 s.
Sensor/Meters Module State Estimator Module	- The time delay for a response is zero. - The configuration defines the set of sampled data (e.g., voltages or currents at specific branches and buses) used for estimating the states of other components in the system.
Protection Systems Module	- Tier 1, Zone 1 protection system on a transmission line will open within 100 ms with a probability of 0.99 after a fault on that line. Therefore, in formula (2), $C = 46.05/s$. - Tier 2, Zone 2 protection system opens and isolates the faulty line if the above condition has been met for 400 ms. Otherwise, the timer will be reset. - Tier 2, Zone 3 protection time delay is defined as 800 ms. - Tier 3, If a transmission line has been overloaded by 200% in current or above for 10 s, then the protection system on this line opens.
HMI Module Operator Module	- The time delay for displaying the information to the operator is zero. - The operator checks the system state every second. When a violation is observed, the operator takes 1 min to perform an action and then waits for 2 min to observe system responses. If the action does not mitigate the violation, the operator will perform another action and observe system responses until no more actions can be performed. During the 1 min that elapses while taking action and the 2 min of observation, the operator will ignore new violations.
Attacker Module	The attacker module is developed to emulate the cyberattacks occurring at system components and their links. The following are the attacker's actions $X_{k,l}$ supported by the current version of the attacker module. In the description, the variable x denotes the original value of a signal from a component, the variable y denotes the actual output value of the signal. I/O Data Blocking Attack discards the original signal/network message being transferred through the target system. Delay Injection introduces a time delay in the transmission of the original signal/network message. Parameter Manipulation and Functional Disabling modifies the original parameter used by a functional module to change the module's behaviors.

constrained system risk R_{sys}^{CS} where the additional constraints in Eqs. (18), (19), and (20) are applied to risk analysis.

4. Case study

4.1. Target system introduction

The proposed DPRA method is demonstrated using the IEEE 14-bus system which was used to provide a realistic, publicly accessible system to the research community. The original IEEE 14-bus test system was developed in 1962 as an approximated representation of the American Electric Power (AEP) system. The model had 14 buses, 5 generators, and 11 loads, and represented power system characteristics of the era. The loads are mostly supported by the generation from Bus 1. The one-line diagram of the modified IEEE 14-bus system is presented in Fig. 3.

For each branch in the grid system, the protection system controls two breakers deployed at each end of the branch. The deployed sensors/meters measure the current and power at each branch, the voltage, frequency, and power of each bus, etc. Table 3 details the elements analyzed in the IEEE 14-bus system.

4.2. Setup of the DPRA modules

As shown in the Fig. 1, several functional modules are required by the DPRA method. This section specifies the configuration of each module for the IEEE 14-bus system case study. Table 4 provides details of the configurations.

According to the guide released by the NIST cyber security working group [37], three main security objectives should be guaranteed to secure Industrial Control Systems (ICSs), which are availability, integrity, and confidentiality. In these objectives, availability denotes the accessibility to data sets and system functions in real time; integrity denotes the consistency of data and commands transferred through the target system and the avoidance of unauthorized alteration of data; confidentiality keeps the sensitive data of the process inaccessible from unauthorized users and assets. Since the violation of availability and integrity will influence the normal operations of industrial control systems, these two objectives are the most important security objectives for ICS [38]. In practice, Denial of Service (DoS) attacks are most commonly used to compromise availability [39], and false data injection (FDI) and Man-in-the-middle (MITM) attacks are most commonly used in compromising integrity [39]. Therefore, the proposed DPRA method

Table 5

Probability values used to describe the uncertainties considered in the case study.

Modules & Links	Probability values for describing uncertainties considered in the case study
System Physics Module Sensors/Meters Module	N/A - One vulnerability was considered, which could be used to conduct FDI attacks. - One attack action was considered, in which the original value was changed to zero. $pv_k = 1$, since only one vulnerability was considered; $pa_{k,l} = 1$, since we assumed that only one attack action $X_{k,l}$ could be conducted; $ps_{k,l}$ was calculated based on the CVSS score; $fa_{k,l}$ a uniform distribution was used; $pf_k = 1$ for the scenarios with physical failures; $pf_k = 0$ for the scenarios without physical failures; ff_k a uniform distribution was used.
Protection Systems	The same as the sensors/meters module, except One attack action was considered, in which the reaction generated by the protection systems was canceled.
State Estimator	The same as the sensors/meters module
HMI	The same as the sensors/meters module
Operator	$ph_k = 1$ for all the reactions; fh_k a uniform distribution was used
Actuator/Breakers	The same as the sensors/meters module
Links	- The same as the sensors/meters module, except - One vulnerability was considered, which could be used to conduct MITM attacks.

focuses on the impacts caused by these types of cyberattacks, and these cyberattacks should be considered first when evaluating the effectiveness of defense strategies. Table 4 details the configuration for each module used by the DPRA method.

The Sensors/Meters module, the Actuators/Breakers module, and the HMI module, perform the function of information transmission between other modules. Specifically, the Sensors/Meters module reads data from the system physics module and send the data to the protection system module and the state estimator module. Extra time delay in the transmissions is the failure mode considered in the proposed method.

The protection systems module acts as the primary protection mechanism in the case of a fault on the branch it protects. A layer of four protection mechanisms has been designed [40,41]. In the protection system, the physical variables (e.g., the measured bus voltage and the measured branch current) are acquired from the network connected to the sensors. Therefore, their values are equal to the corresponding cyber variables defined in the links. The cyberattacks performed by the attacker modules may change the values of the cyber variables and finally impact the values of these physical variables.

The Operator Module represents possible operator actions under various situations. It is implemented based on operational rules identified from unstructured interviews with operators at a vertically integrated utility that serves the Balancing Authority (BA) and transmission service area. In this paper, the Operator Module is simplified as a single step due to the lack of operational data which estimates the operator's behaviors and response time for different system configurations. However, the decision-making and execution response time variability is high enough that considering those steps as a single process may not significantly impact the simulation results. Each cognitive process often happens in the milliseconds time range [42], and a motor task often happens in the seconds time range. In electric grid operation, the decision-making and execution response time vary in the minutes time range.

In addition, the parameters and thresholds mentioned in Table 4 can be modified dynamically by the attacker module (except the ones related to the System Physics Module, the Fault Injection Module, and the State Estimator Module) to emulate the attacker's actions for manipulating system behaviors.

Table 5 shows the probability values used to describe the uncertainties considered in the case study. To accelerate the risk assessment process, we predefined several constraints for those modules used by the DPRA method. Specifically, we assumed that at least one vulnerability exists in the modules which use digital systems and are vulnerable to cyberattacks. Also, we used uniform distribution for the occurrence of cyberattacks. We ensured that at least one physical failure would occur in the scenarios with physical failures.

4.3. Simulated scenarios

Three types of scenarios are analyzed for the case study system. These different types are intended to cover a broad range of attack scenarios with varying complexity.

- **Scenario Type I (Single Attack Scenarios):** In the single cyber-attack scenarios, the attack occurs randomly at various locations, including the voltage and current meters at the substation of bus 5, the field bus connecting the meters, and the protection systems at the substation of bus 5, the network connecting the meters and the state estimator, the network connecting the state estimator and the HMI, the HMI at the operation center.
- **Scenario Type II (Double Attack Scenarios):** Two attacks occur in each scenario in the category of double cyberattack scenarios. One of the two cyberattacks targets the protection systems at the substation of bus 5, in which the breaker at branch 4–5 is compromised and disabled, i.e., the breaker cannot open even if the calculated Zone 2 and Zone 3 impedance values are below their respective thresholds.

The other cyberattack is similar to the one in the single attack scenarios, which will occur randomly at various locations.

- **Scenario Type III (Double Attack Scenarios with A Physical Fault):** The third category of scenarios includes two cyberattacks similar to the ones in double cyberattack scenarios. Besides, a “three-phase to ground” fault, which is the most severe type of fault (i.e., causing the most damage), will be injected at the 50% location of branch 4–5 in each scenario.

The success probability of cyberattacks $ps_{k,l}$ in each type of attack scenario needs to be defined for each scenario. The CVSS score uses three factors to calculate the likelihood that a component may be attacked successfully:

$$Exploitability (Weight) = 20 * AccessComplexity * Authentication * AccessVector$$

Different values of the three factors correspond to different levels of security of a component. For Access Complexity, three levels are considered, and the weight for each level is as follows: High (H): 0.35; Medium (M): 0.61; Low (L): 0.71. For Authentication, the levels of security and the corresponding weights are: Requires no authentication (N): 0.704; Requires single instance of authentication (S): 0.56; Requires multiple instances of authentication (M): 0.45. For Access Vector, the levels of security and the corresponding weights are: Requires local access (L): 0.395; Local Network accessible (A): 0.646; Network accessible (N): 1. Based on the level of security for each type of cyberattack, the overall exploitability or weight can be calculated. The success rate of each attack $ps_{k,l}$ relative to the one for default system configuration is obtained by calculating the weight ratios of different system configurations. Assume that the success rate of the cyber attacks applied to the system without defense strategies, aka the default system configuration, is 100%, which is used as a baseline. Then, the success rate of the cyberattacks targeting the system with defense strategies can be calculated by the following formula.

$$\gamma = \frac{Weight\ of\ the\ configuration\ with\ defense\ strategies}{Weight\ of\ the\ configuration\ without\ defense\ strategies} \times 100\%$$

For example, if the weight of the default configuration is 4.413 and the weight of the configuration with enhanced meters is 2.035, then the success rate of the cyberattack on the later configuration is $\frac{2.035}{4.413} \times 100\% = 46.11\%$. Table 6 shows the calculated weights and success rates of the cyberattacks targeting various system configurations.

In the case study, three defense strategies are evaluated, including 1) enhancing cybersecurity for the smart meters, 2) increasing the difficulty of accessing the meters' buses, and 3) securing the HMI at the operator center.

We understand that directly using Common Vulnerability Scoring System (CVSS) scores [43] for risk assessment has several shortcomings since CVSS heavily relies on expert judgment and does not consider system dynamics (e.g., the interactions between system components). Therefore, in the proposed method, we reused the CVSS scores only to differentiate between the level of difficulty to successfully conduct a cyberattack.

The IEEE 14-bus system is considered as failed if one of the following criteria is satisfied. System failure refers to the situation when the system or a part of the system collapses due to major outages or cascading failures [44]. In this research, we enumerate the scenarios that could lead to system failure and investigate detailed consequences. The scenarios are categorized as the following three groups of Criteria so that case studies can be performed in a trackable fashion. Since the IEEE 14-bus system is relatively small and Bus 1 supplies a large percentage of power, some cases with two concurrent line outages (N-2 contingencies) could lead to system failure, as listed in Criterion I.

Table 6

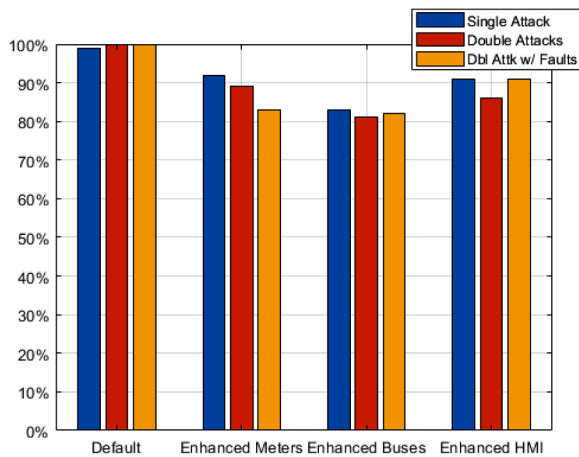
wt and success rates of the cyberattacks targeting various system configurations.

Configurations	Cyberattack Positions	Access Complexity	Authen-tication	Access Vector	Weight (E)	Success Rate
Default	Meters compromised	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	MITM between the meters and protection systems	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	MITM between the meters and the state estimator	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	MITM between the state estimator and the HMI	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	HMI's compromised	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
Enhanced Meters	Meters compromised	0.35 (H)	0.45 (M)	0.646 (A)	2.035	46.11%
	MITM between the meters and protection systems	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	MITM between the meters and the state estimator	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	MITM between the state estimator and the HMI	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	HMI's compromised	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
Enhanced Meter Buses	Meters compromised	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	MITM between the meters and protection systems	0.35 (H)	0.56 (S)	0.646 (A)	2.532	57.38%
	MITM between the meters and the state estimator	0.35 (H)	0.56 (S)	0.646 (A)	2.532	57.38%
	MITM between the state estimator and the HMI	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	HMI's compromised	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
Enhanced HMI	Meters compromised	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	MITM between the meters and protection systems	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	MITM between the meters and the state estimator	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	MITM between the state estimator and the HMI	0.61 (M)	0.56 (S)	0.646 (A)	4.413	100%
	HMI's compromised	0.35 (H)	0.45 (M)	0.646 (A)	2.035	46.11%

Table 7

Failure cases with two branches open.

Branch Combinations	
Branch 1–2 and Branch 1–5	Branch 7–8 and Branch 4–7
Branch 1–2 and Branch 2–5	Branch 7–8 and Branch 4–9
Branch 1–2 and Branch 4–5	Branch 7–8 and Branch 7–9
Branch 2–4 and Branch 4–5	Branch 4–7 and Branch 7–9
Branch 7–8 and Branch 2–3	

**Fig. 4.** Conditional Probabilities of System Failures for Each Defense Strategy.**Table 8**

Conditional failure probabilities of various system configurations under various attack scenarios.

Attack Scenarios	Configuration	Criterion I	Criterion II	Criterion III	System conditional failure probability
Single Attack	Default	99.00%	33.00%	33.00%	99.00%
	Enhanced Meters	92.00%	25.00%	25.00%	92.00%
	Enhanced Meter Buses	83.00%	26.00%	26.00%	83.00%
	Enhanced HMI	91.00%	33.00%	33.00%	91.00%
Double Attack	Default	100.00%	48.00%	48.00%	100.00%
	Enhanced Meters	89.00%	38.00%	38.00%	89.00%
	Enhanced Meter Buses	81.00%	35.00%	35.00%	81.00%
	Enhanced HMI	86.00%	48.00%	48.00%	86.00%
Dbl Attk w/ Fault	Default	39.00%	100.00%	42.00%	100.00%
	Enhanced Meters	22.00%	83.00%	25.00%	83.00%
	Enhanced Meter Buses	33.33%	82.83%	35.35%	82.83%
	Enhanced HMI	39.00%	91.00%	40.00%	91.00%

- **Criterion I:** The system fails if any two branches listed in Table 7 are open. The concurrent opening of these branches causes isolated load or generation pockets, severe under/over voltages, line overloading, or outright system collapse.
- **Criterion II:** The system fails if any three of the following branches are open concurrently: {Branch 1–2, Branch 1–5, Branch 2–4, Branch 2–5, Branch 3–4, Branch 4–5, Branch 5–6, Branch 4–7, Branch 4–9, Branch 7–8, Branch 7–9}.
- **Criterion III:** For the scope of this research, the operating system state is considered as failed if the total load supplied is reduced to 60% or less of the nominal amount due to equipment outages on the system.

4.4. Simulation results

Fig. 4 compares the probabilities of system failures given each defense strategy for the three types of scenarios considered in Section 4.3, i.e., single attacks, double attacks, and double attacks with physical faults. It is worth noting that these probabilities are conditional probabilities, assuming the occurrence of one or two cyberattacks. Table 8 shows the conditional failure probabilities of various system configurations under the predefined attack scenarios for the three aforementioned criteria for system failure. Each system conditional failure probability in the last column of Table 8 denotes the conditional probability that any of the three failure criteria is satisfied.

In the case study, 100 simulations are conducted per configuration per attack scenario (2400 simulations for all combinations of attacks and configurations). In each simulation, a time window of 600 s was considered. Based on the system failure probability, enhancing the

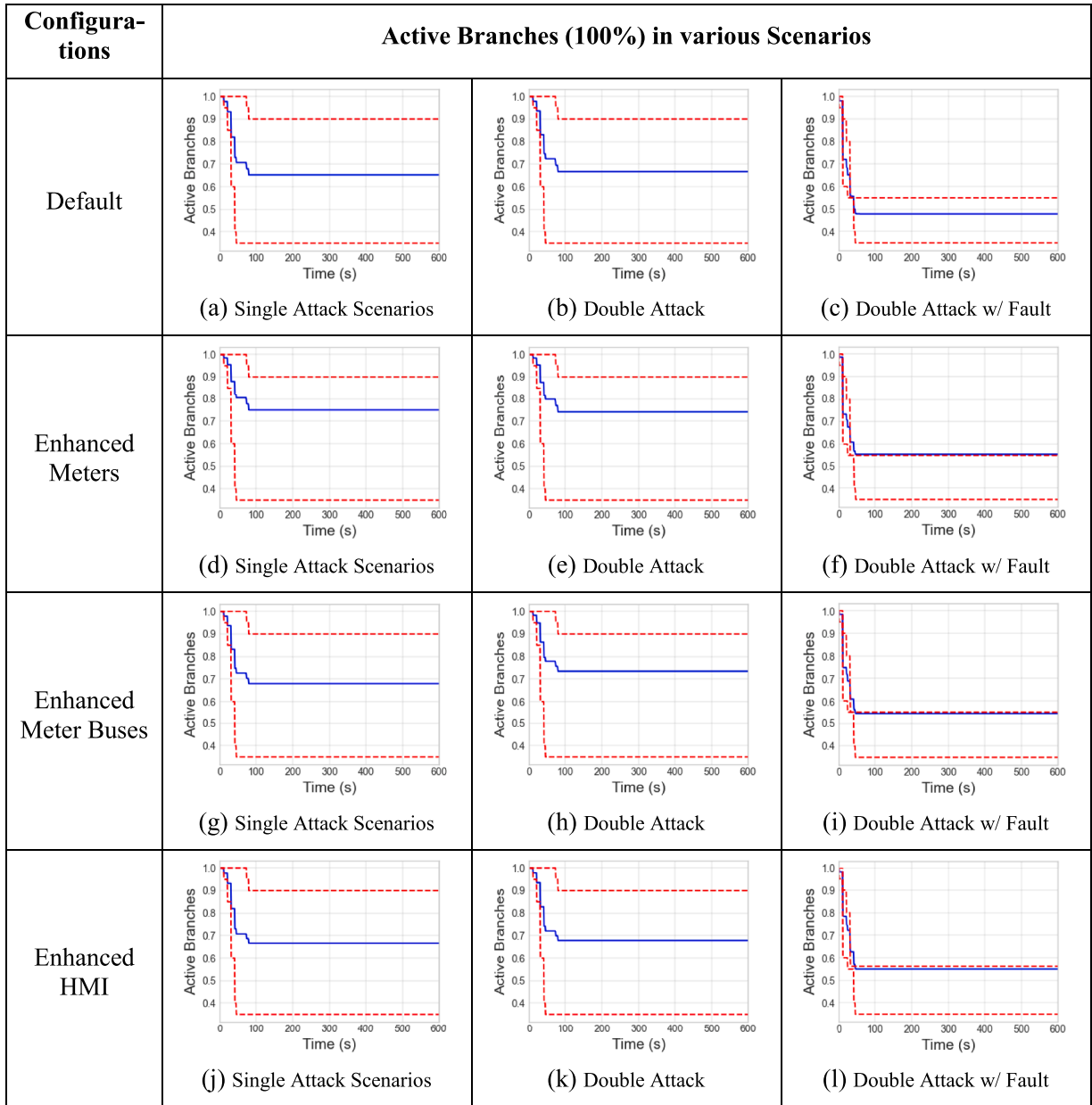


Fig. 5. Experiment results from the IEEE 14-Bus system, solid line: the mean values (blue), dashed line: the 25% quantiles (lower, red), the 75% quantiles (higher, red).

meter's buses is the most effective defense strategy since it reduces the system failure probabilities the most in single-attack scenarios (from 99% to 83%), double-attack scenarios (from 100% to 81%), and scenarios with two attacks and a physical fault (from 100% to 82.83%). However, these reductions vary with specific failure criteria used in obtaining the system failure probabilities. For example, if failure criterion III (i.e., the loads' drop criterion) is considered only, then the most effective defense strategy in the case of single attack scenarios is enhancing the security of meters since the failure ratio decreases from 33% to 25%. In this case, enhancing the HMI does not impact the failure probability; hence it is the least effective defense strategy.

Moreover, observing the time history of the number of active branches (i.e., the branches that remain closed) can help acquire more detailed information about the system evolution since the larger the number of active branches, the larger the capability of the grid for transferring electric power. Fig. 5 illustrates the average number of active branches of the system (the solid blue lines), shown as the mean percentage of the closed branches, and the 25% and 75% quantiles (the

lower and higher red dashed lines). These are obtained based on the 100 simulations for each type of scenario and each configuration. The figures are grouped into rows for different system configurations. The figures from top to bottom illustrate the results for the system configurations of Default, Enhanced Meters, Enhanced Meter Buses, and Enhanced HMI, respectively. In each row, the first figure displays the results of the single attack scenarios. The second and third figures are the results of double attack scenarios and scenarios with double attacks and physical faults, respectively. We can observe that the curves of double attacks with physical faults scenarios (Fig. 5(c), (f), (i), (l)) drop faster than the curves of single attack and double attacks since the situations with both cyber and physical events are more severe in which faults and cyberattacks coincide. Specifically, in Fig. 5(a), the active branches dropped from 100% to around 65% in around 80 s. Fig. 5(c) shows that the ratio drops more quickly from 100% to less than 50% in 50 s. We can also observe that the differences between the 25% and 75% quantiles in the figures of single attack scenarios and double attack scenarios are much larger than the ones in the figures of double attack with fault scenarios. This implies

that large uncertainties exist in single-attack and double-attack scenarios.

By applying the Enhanced Meters defense, the average ratio of closed branches in single attack scenarios at 600 s remains above 75% (see Fig. 5(d)), which is higher than the default configuration. Also, the ratios of closed branches in the scenarios of double attacks and double attacks with faults are 75% and 55%, respectively. Both ratios are higher than the ones in the case of the default configuration, in which the ratio under the single attack scenario is 66%, and the ratio under the double attack with faults scenario is 48%. Improvements can also be observed from the Enhanced Meter Buses and Enhanced HMI configurations. The figures imply that if we use the average number of branches remaining closed after 600 s as a criterion, the Enhanced Meter is the most effective defense strategy.

The output of the method includes the system failure rates when applying different defense strategies to evaluate the effectiveness of these strategies. System designers, especially the designers of defense mechanisms, can use the results of this method to evaluate the effectiveness of their defense strategies. Decisions can be made by referring to the reduction of system risks when applying the defense mechanisms at different locations. For example, using the default configuration as a benchmark, applying a defense strategy at the meters can decrease the success probability of compromising a meter from 100% to 46.11% (see Table 6). The decreased probability can also be acquired by using historical data or experiments related to similar defense mechanisms. Then, the proposed method can automatically evaluate the impact of such improvement on the system risk. For comparing the effectiveness of various defense strategies, a similar defense mechanism can be applied to the HMI, in which the success probability of compromising an HMI decreases from 100% to 46.11% (see the last row in Table 6). Then, the proposed method can also evaluate such improvement to the system risk. Finally, by comparing the calculated risk for the two different defense strategies, a more efficient defense strategy can be identified. In the case study, the defense strategy applied to the HMI is more efficient than the one applied to the meters in defending double attacks since the former reduced the system risk (i.e., the probability of system failure) from 100% to 89%, while the latter reduced the system risk from 100% to 86%. However, from Table 8, we can observe that for mitigating the impacts of both cyberattacks and physical faults, the defense strategy applied to the meters is better than the one applied to the HMI. Similarly, the Enhanced Bus Meters strategy has the best performance in defending against single attacks since the system risk is reduced from 99% to 83%, compared with the reduction of 99% to 92% in the case of the Enhanced Bus strategy and 99% to 91% in the case of the Enhanced HMI strategy. In the current method, we use quantified system risk for evaluating the effectiveness of various defense strategies. Discrete security levels can also be defined for specific purposes. For example, Security Level I can be defined as the probability of system failure needing to be lower than 50% even if a cyberattack is being conducted. Then, using the same context, Level II can be defined as requiring the probability of system failure to be lower than 20%, and Level III can be defined as requiring the probability of system failure to be lower than 5%. In addition, in making a decision, decision makers will need to consider the costs associated to the deployment of such defense strategies. For the case study at hand, the decision maker might consider studying the combination of the defense strategies exemplified as well.

5. Discussion

In the case study section, we used the proposed DPRA method to evaluate the effectiveness of various defense strategies for three types of attack scenarios instead of performing a complete risk assessment for grid systems. We introduced several constraints to accelerate the process of analysis to demonstrate the advantages of the proposed method. In the case study, we simulated the trajectory of the system for 600 s. Due to the fast evolution of the states of the grid, assessing risks for minutes

will generate valuable results (e.g., the grid will fail in minutes in a cascading failure). When analyzing the effectiveness of cyber defense mechanisms, the time horizon of hours and days may not be applicable or necessary since a delay in the response of the defense mechanisms on the order of seconds will significantly impact the consequences. A delay in hours or days is not acceptable for cyber defense mechanisms. In the case study section, we only used brutal cyberattacks, which will cause cascading failures in seconds. However, the proposed method can also be used to analyze advanced persistent attacks and simulate them for hours and days with enough computer memory space or by introducing a save and restart option which is not currently available.

The proposed method can perform a probabilistic risk assessment of a broader scope by introducing the probabilities of occurrence of various cyberattack scenarios and removing the constraints introduced in the case study section. Generally, it is difficult to predict the likelihood of cyberattacks, but some data sources can be referenced to approximate the likelihood of cyberattacks [45–47]. This paper does not discuss the likelihood of cyberattacks. Instead, it defines the success probability of cyberattacks in a specific attack scenario in which a specific defense strategy is deployed. Therefore, the outcome probability is a conditional probability in which we assume one or more vulnerabilities exist in the target system and a cyberattack occurs. We used conditional probabilities to evaluate the effectiveness of defense strategies.

Also, in this paper, we selected some commonly used cyberattacks to demonstrate the effectiveness of the proposed method in evaluating the effectiveness of defense strategies. The methodology of threat modeling, e.g., STRIDE [48,49], can be helpful for defining the likelihood of cyberattacks and possible vulnerabilities residing within the target system. Threat modeling methodologies can be a future research direction to improve the accuracy of the risk predictions for cyberattacks.

The proposed method can be adjusted to analyze zero-day vulnerabilities and black-swan events. A zero-day vulnerability is a vulnerability in a system that has been disclosed but is not fixed yet. A black-swan event is an event that has never happened before. All these events may cause catastrophic consequences to the target system. In the case of zero-day vulnerabilities, to analyze system risk with such vulnerabilities, we can first identify the target components of such vulnerabilities in the system and then assign a high success rate (e.g., 99%) to the cyberattacks using such vulnerabilities.

In the case of black-swan events, the proposed method provides flexible expressions that allow system analysts to define the actions of attackers, such as modifying any signals handled by functional modules and transferred by links. System analysts can define various attacker actions, including actions that may have never occurred. A uniform distribution over possible locations (e.g., sensors, HMI) and possible action types can be used to model the occurrence of these events and can support the performance of the conservative analysis.

For applying the proposed method to other types of cyber-physical systems, the following modules or functions can be updated and extended.

- The attacker module can be reused to simulate cyberattacks. Technically, the attacker module does not perform “attacks” on the other functional modules. Instead, it emulates the behaviors of attackers when they successfully compromise a system component. Generally, such behaviors include the manipulations of data used or generated by the compromised component. Therefore, these behaviors can be directly applied to the functional modules that are designed for other complex systems.
- The fault generation module can be reconfigured with different distributions to simulate physical faults that may occur in other types of systems. Since the fault generation module requires the API provided by the simulators used by the system physics module to inject physical faults into the simulated system, the interfaces calling the APIs need to be updated if another simulator is used. The other parts of the fault generation module can be reused.

- The sensor/meters module, the actuators/breakers module and the HMI module can be reused and reconfigured. These modules use configuration files to manage the data handled by the module.

6. Conclusion

This paper introduces a dynamic probabilistic risk assessment method designed to perform risk analysis for electric grid systems under cyberattacks. As a case study, this paper uses the proposed method to evaluate the effectiveness of different defense strategies that can be applied to various system components and links. The proposed method provides a way of evaluating the reduction in cybersecurity risk due to the introduction of defense mechanisms. Therefore, it can be used by system designers to determine the most effective defense strategies for enhancing cybersecurity of an electric grid. The following summarizes the findings of the paper:

- The proposed DPRA method can model and simulate the behaviors of critical entities (e.g., meters, protection systems, operators) involved in system operations. Common attacker actions (e.g., changing the original signals in a data bus) can be defined and executed during the simulation. Also, the operator's responses to some classical accidental events are modeled. These modules can be further improved and reused in the analysis for other types of systems.
- The proposed method accepts various constraints that can be defined before the simulation. These constraints can guide the method to analyze the scenarios with various occurrence probabilities and prune the impossible scenarios so as to accelerate the risk assessment process. In this paper, the vulnerabilities, the probabilities of exploiting these vulnerabilities, the probabilities of successfully conducting a cyberattack by exploiting such vulnerabilities, etc., are introduced as the constraints for evaluating the effectiveness of defense strategies.
- In this paper, the proposed method was used to evaluate the effectiveness of defense strategies that can resist attackers' intrusions. The results from the analyses for different defense strategies, such as the system configuration with security-enhanced meters and buses, were obtained, including failure probabilities, loss of loads, opened branches, etc. According to the obtained results, we can infer that using single position defense for the target system is not sufficient since the system risk under cyberattacks decreased only less than 20%. More sophisticated defense mechanisms are required.

Future work includes: 1) Improving the modeling of attack behaviors in the attacker module to simulate complex cyberattack scenarios. 2) Enhancing the operator module to reflect complex operators' actions in the real world. 3) Adding new simulated components (e.g., the firewall, the software-defined networking components) that may be required for analyzing complex system architectures. 4) Threat models, such as STRIDE, will be used to determine and improve the occurrence probability of various cyberattacks. 5) Integrating the proposed method with multiple grid simulators.

CRediT authorship contribution statement

Xiaoxu Diao: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Yunfei Zhao:** Conceptualization, Methodology, Software, Writing – review & editing. **Carol Smidts:** Conceptualization, Methodology, Writing – review & editing. **Pavan Kumar Vaddi:** . **Ruixuan Li:** Resources, Writing – original draft. **Hangtian Lei:** Software, Writing – original draft. **Yacine Chakhchoukh:** Software, Writing – original draft. **Brian Johnson:** Software, Supervision. **Katya Le Blanc:** Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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